MATHS MANTHAN

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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 09 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M, 3M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If the nth term of an AP is $a_n = 3n + 4$, then its first term a_1 is: [1]

(A) 3 (B) 4 (C) 7 (D) 10

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 7** [0.5 Mark]

We evaluate a_n at $n = 1$:

Formula / Theorem Used: $a_1 = a_n|_{n=1}$

Justification: [0.5 Mark]

$a_1 = 3(1) + 4 = 3 + 4 = \mathbf{7}$.

Q2. If the nth term of an AP is $a_n = 5 - 2n$, then its common difference d is: [1]

(A) 5 (B) 2 (C) -2 (D) 3

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) -2** [0.5 Mark]

We use the common difference formula:

Formula / Theorem Used: $d = a_2 - a_1$

Justification: [0.5 Mark]

$a_1 = 5 - 2(1) = 3$ and $a_2 = 5 - 2(2) = 1$, so $d = 1 - 3 = \mathbf{-2}$.

Q3. The 10th term of the AP $2, 7, 12, \ldots$ is: [1]

(A) 47 (B) 42 (C) 52 (D) 37

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 47** [0.5 Mark]

Given: $a = 2$, $d = 7 - 2 = 5$, $n = 10$.

Formula / Theorem Used: $a_n = a + (n - 1)d$

Justification: [0.5 Mark]

$a_{10} = 2 + (10 - 1)(5) = 2 + 9 \times 5 = 2 + 45 = \mathbf{47}$.

Q4. If the nth term of an AP is $a_n = 7 - 4n$, then its common difference is: [1]

(A) 7 (B) 4 (C) -4 (D) 3

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: (C) -4 [0.5 Mark]

We evaluate consecutive terms:

Formula / Theorem Used: $d = a_2 - a_1$

Justification: [0.5 Mark]

$a_1 = 7 - 4(1) = 3$ and $a_2 = 7 - 4(2) = -1 \implies d = -1 - 3 = \mathbf{-4}$.

Q5. **Assertion (A):** In an AP with common difference d, $a_{20} - a_{10} = 10d$.
Reason (R): The nth term of an AP is $a_n = a + (n-1)d$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: (A) Both (A) and (R) are true, and (R) explains (A). [0.5 Mark]

Applying the formula in Reason (R):

Formula / Theorem Used: $a_m - a_n = [a + (m-1)d] - [a + (n-1)d] = (m-n)d$

Justification: [0.5 Mark]

Reason (R) states the true general term formula. Setting $m = 20, n = 10$ yields $a_{20} - a_{10} = (a + 19d) - (a + 9d) = 10d$. Hence both are true and (R) explains (A).

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Q6. Find the value of k for which k^2+4k+8, $2k^2+3k+6$, and $3k^2+4k+4$ are three consecutive terms of an AP. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Three consecutive terms $a = k^2+4k+8$, $b = 2k^2+3k+6$, $c = 3k^2+4k+4$.

Formula / Theorem Used: If a, b, c are in AP, then $2b = a + c$

Step 1: Setting Up the Condition [1 Mark]

$$2(2k^2+3k+6) = (k^2+4k+8) + (3k^2+4k+4)$$

Step 2: Simplifying and Solving for k [1 Mark]

$$4k^2+6k+12 = 4k^2+8k+12 \implies 6k = 8k \implies 8k-6k = 0 \implies 2k = 0 \implies \boxed{\mathbf{k = 0}}$$

(Verification: at $k = 0$, the three terms are $8, 6, 4$, forming an AP with $d = -2$.)

Q7. If the 7th term of an AP is $\frac{1}{9}$ and its 9th term is $\frac{1}{7}$, find its 63rd term. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $a_7 = a + 6d = \frac{1}{9}$ — (i) and $a_9 = a + 8d = \frac{1}{7}$ — (ii).

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Finding d and a [1 Mark]

Subtracting (i) from (ii):

$$2d = \frac{1}{7} - \frac{1}{9} = \frac{9-7}{63} = \frac{2}{63} \implies d = \frac{1}{63}$$

Substituting $d = \frac{1}{63}$ into (i): $a + 6\left(\frac{1}{63}\right) = \frac{1}{9} \implies a = \frac{7}{63} - \frac{6}{63} = \frac{1}{63}$.

Step 2: Computing a_{63} [1 Mark]

$$a_{63} = a + 62d = \frac{1}{63} + 62\left(\frac{1}{63}\right) = \frac{1+62}{63} = \frac{63}{63} = \boxed{1}$$

Q8. Find the middle term of the AP $6, 13, 20, \ldots, 216$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: $a = 6$, $d = 13 - 6 = 7$, last term $a_n = 216$.

Formula / Theorem Used: $a_n = a + (n-1)d$ and Middle index (odd n) $= \dfrac{n+1}{2}$

Step 1: Finding the Number of Terms (n) [1.5 Marks]

$$6 + (n-1)(7) = 216 \implies 7(n-1) = 216 - 6 = 210 \implies n - 1 = \frac{210}{7} = 30 \implies n = 31$$

Step 2: Finding the Middle Term [1.5 Marks]

Since $n = 31$ (odd), the middle term is the $\left(\frac{31+1}{2}\right)^{\text{th}} = 16^{\text{th}}$ term:

$$a_{16} = 6 + (16-1)(7) = 6 + 15 \times 7 = 6 + 105 = \boxed{111}$$

Q9. Check whether -150 is a term of the AP $11, 8, 5, 2, \ldots$; justify your answer. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: AP is $11, 8, 5, 2, \ldots$ with $a = 11$ and $d = 8 - 11 = -3$.

We use the general term test:

Formula / Theorem Used: $a_n = a + (n-1)d$, where index n must be a positive integer ($n \in \mathbb{N}$).

Step 1: Assuming $a_n = -150$ and Solving for n [2 Marks]

$$11 + (n-1)(-3) = -150 \implies -3(n-1) = -150 - 11 = -161 \implies n - 1 = \frac{161}{3}$$

$$n = \frac{161}{3} + 1 = \frac{161+3}{3} = \frac{164}{3} = 54\frac{2}{3}$$

Step 2: Concluding Validity [1 Mark]

Since $n = \frac{164}{3}$ is not a positive integer ($\frac{164}{3} \notin \mathbb{N}$), we conclude:

No, -150 is NOT a term of the given AP.

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Q10. The ratio of the sums of the first m terms and the first n terms of an AP is $m^2 : n^2$. Show that the ratio of its mth and n^{th} terms is $(2m-1) : (2n-1)$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let first term be a and common difference be d. Given $\dfrac{S_m}{S_n} = \dfrac{m^2}{n^2}$.

Formula / Theorem Used: $S_k = \dfrac{k}{2}[2a + (k-1)d]$ and $a_k = a + (k-1)d$

Step 1: Simplifying the Ratio of Sums [1.5 Marks]

$$\frac{\frac{m}{2}[2a + (m-1)d]}{\frac{n}{2}[2a + (n-1)d]} = \frac{m^2}{n^2} \implies \frac{m}{n} \cdot \frac{2a + (m-1)d}{2a + (n-1)d} = \frac{m^2}{n^2} \implies \frac{2a + (m-1)d}{2a + (n-1)d} = \frac{m}{n}$$

Step 2: Cross-Multiplying to Express d in Terms of a [2 Marks]

$$n[2a + (m-1)d] = m[2a + (n-1)d]$$

$$2an + mnd - nd = 2am + mnd - md \implies 2a(n-m) = d(n-m)$$

For $m \neq n$ (so $n - m \neq 0$), dividing by $(n-m)$ gives $d = 2a$. (For $m = n$, $\frac{a_m}{a_n} = 1 = \frac{2m-1}{2n-1}$ trivially.)

Step 3: Evaluating the Ratio $a_m : a_n$ [1.5 Marks]

$$\frac{a_m}{a_n} = \frac{a + (m-1)d}{a + (n-1)d} = \frac{a + (m-1)(2a)}{a + (n-1)(2a)} = \frac{a(1 + 2m - 2)}{a(1 + 2n - 2)} = \frac{2m-1}{2n-1}$$

$$\boxed{\mathbf{a_m : a_n = (2m-1) : (2n-1)} \quad \text{[Hence Proved]}}$$

Q11. A contract on a construction job specifies a penalty for delay of completion beyond a certain date as follows: Rs. 200 for the first day, Rs. 250 for the second day, Rs. 300 for the third day, etc., the penalty for each succeeding day being Rs. 50 more than for the preceding day. How much money does the contractor have to pay as penalty if he delays the work by 30 days? **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Modelling Daily Penalties as an AP [1.5 Marks]

The daily penalties (in Rs.) are $200, 250, 300, \ldots$ for 30 days.
Since consecutive increases are constant, this forms an AP with $a = 200$, $d = 250 - 200 = 50$, and $n = 30$.

Formula / Theorem Used: $S_n = \dfrac{n}{2}[2a + (n-1)d]$

Step 2: Substituting into the Sum Formula [2 Marks]

$$S_{30} = \frac{30}{2}[2(200) + (30-1)(50)] = 15[400 + 29 \times 50]$$

Step 3: Arithmetic Evaluation and Final Statement [1.5 Marks]

$$S_{30} = 15(400 + 1{,}450) = 15 \times 1{,}850 = 27{,}750$$

The contractor has to pay a total penalty of $\boxed{\text{Rs. } \mathbf{27{,}750}}$.

Q12. CASE STUDY: Systematic Monthly Savings Plan (Aarav)

To build an emergency education fund, Aarav deposits Rs. 1,500 in Month 1 and increases his monthly deposit uniformly by Rs. 250 every subsequent month over a period of 36 months (3 years). [5]

(i) Calculate Aarav's deposit during the 12th month. [1]

(ii) Determine the amount deposited during the 24th month. [1]

(iii) Compute the total cumulative savings deposited by Aarav across all 36 months. [3]

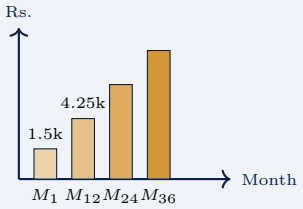
— **OR** —

In which month will Aarav's monthly deposit reach exactly Rs. 10,000? [3]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Extracted Scenario Data: Monthly deposits form the AP 1,500, 1,750, 2,000, ... with $a = 1{,}500$, $d = 250$, $n = 36$.

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$



Sub-part (i): Deposit in the 12th Month [1 Mark]

$a_{12} = 1{,}500 + (12-1)(250) = 1{,}500 + 11 \times 250 = 1{,}500 + 2{,}750 =$ **Rs. 4,250**.

Sub-part (ii): Deposit in the 24th Month [1 Mark]

$a_{24} = 1{,}500 + (24-1)(250) = 1{,}500 + 23 \times 250 = 1{,}500 + 5{,}750 =$ **Rs. 7,250**.

Sub-part (iii): Total Cumulative Savings Over 36 Months [3 Marks]

$$S_{36} = \frac{36}{2}[2(1{,}500) + (36-1)(250)] = 18(3{,}000 + 35 \times 250)$$
$$= 18(3{,}000 + 8{,}750) = 18 \times 11{,}750 = \textbf{Rs. 2,11,500}$$

Sub-part (iii) [OR Option]: Month When Monthly Deposit Reaches Rs. 10,000 [3 Marks]

Let $a_n = 10{,}000$:

$$1{,}500 + (n-1)(250) = 10{,}000 \implies 250(n-1) = 8{,}500 \implies n - 1 = \frac{8{,}500}{250} = 34 \implies n = \textbf{35}^{\text{th}} \text{ month}$$

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